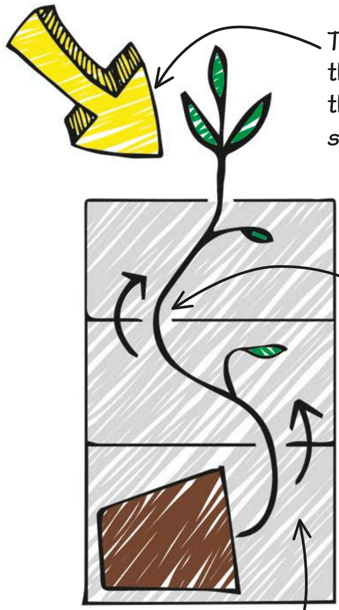
HOW IT WORKS

Plants use the energy in sunlight to make their own food. In a process called photosynthesis, they soak up the sun's energy with their leaves and use it to turn water from the soil and carbon dioxide from the air into glucose, a type of sugar. Glucose provides plants with fuel. It is no surprise, then, that plants have evolved a way to grow toward light, so they can sunbathe as much as possible. It's all down to chemicals called auxins. The more auxins in a particular part of the plant, the faster that part will grow. Light destroys auxins, so there will be fewer of them in the side of the stem that receives most light. The shaded side of the plant contains more auxins, so that side grows more quickly. This causes the plant to bend toward the light.



REAL WORLD SCIENCE
PLANT GROWTH

A young sunflower turns its head from east to west during the day, following the sun. At night, it turns back to face east again, ready for sunrise the next morning. Both movements are caused by changing levels of auxins inside the plant. Once the sunflower's head has developed, the plant stops following the sun, usually settling with its head facing east.

PHOTOSYNTHESIS

During daylight, a tree makes glucose by photosynthesis. At night, it uses the nourishing glucose to stay alive and to keep growing.